

Revised Generation Interconnection System Impact Study Report

For Queue Project Q23-2302

Bluebonnet 230 kV
October 23, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2302. The Interconnection Customer proposes a 500.0 MW battery energy storage facility interconnecting at the Bluebonnet 230 kV substation in the Indiana area. The request was received on March 23, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 529 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,580,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: BLUEBONNET-JONES CHAPEL 230 kV
Contingency: ROBISON PARK-RUSSELL FORK 230 kV Line
Loading: 131.5% (Rating: 1,701 MVA)
DFAX: 60.8%
Pre-project loading: 113.7%
2. Facility: BREED-MCNEIL 230 kV
Contingency: RICHLAND-DEKALB 500 kV Line
Loading: 111.0% (Rating: 2,568 MVA)
DFAX: 23.8%
Pre-project loading: 106.3%
3. Facility: SLOCUM RD-MCNEIL 230 kV
Contingency: RICHLAND-DEKALB 500 kV line
Loading: 111.0% (Rating: 2,550 MVA)
DFAX: 23.8%
Pre-project loading: 106.3%
4. Facility: DEKALB-SLOCUM RD 230 kV
Contingency: DEKALB-RICHLAND 500 kV
Loading: 110.8% (Rating: 2,585 MVA)
DFAX: 23.8%
Pre-project loading: 106.2%
5. Facility: SIDNEY A MURRAY JR HYDROELECTRIC-24TH & GARDENDALE 230 kV
Contingency: RICHLAND SUBSTATION-DEKALB 500 kV
Loading: 108.6% (Rating: 1,060 MVA)
DFAX: 8.5%
Pre-project loading: 104.6%

6. Facility: RICHLAND SUBSTATION 500/230kV Transformer
Contingency: TROY ENERGY LLC-RICHLAND SUBSTATION 500 kV
Loading: 105.4% (Rating: 3,175 MVA)
DFAX: 28.9%
Pre-project loading: 100.8%
7. Facility: MINNEHAHA-NAPOLEON PEAKING STA SUBSTATION 230 kV
Contingency: TROY ENERGY LLC-RICHLAND 500 kV Line
Loading: 104.0% (Rating: 1,575 MVA)
DFAX: 11.6%
Pre-project loading: 100.3%
8. Facility: RITTER PARK-SUGAR HILL 345 kV
Contingency: HILLTOP-PINEY CREEK PROJECT 500 kV line
Loading: 103.6% (Rating: 2,427 MVA)
DFAX: 9.8%
Pre-project loading: 101.6%
9. Facility: RICHLAND-DEKALB 500 kV
Contingency: SLOCUM RD SUBSTATION-DEKALB 230 kV
Loading: 102.9% (Rating: 5,511 MVA)
DFAX: 49.1%
Pre-project loading: 98.4%
10. Facility: LIGHTHOUSE HILL-STEWART MTN DAM 345 kV Line
Contingency: PINEY CRK PROJECT-HILLTOP 500 kV
Loading: 100.9% (Rating: 1,706 MVA)
DFAX: 7.0%
Pre-project loading: 98.9%
11. Facility: DEKALB 500 kV/1 kV TX
Contingency: JONES CHPEL-DEKALB 500 kV
Loading: 100.9% (Rating: 3,072 MVA)
DFAX: 26.0%
Pre-project loading: 96.6%
12. Facility: DEKALB 230/1kV TX
Contingency: DEKALB-JOES CHAPEL 500 kV
Loading: 100.9% (Rating: 3,072 MVA)
DFAX: 26.0%
Pre-project loading: 96.6%
13. Facility: ATASCOCITA-MAPLEST 230 kV
Contingency: RAVENNA-EL PASO 500 kV
Loading: 100.3% (Rating: 1,011 MVA)
DFAX: 5.0%
Pre-project loading: 97.9%
14. Facility: LEETONIA-MAPLE ST 230 kV L
Contingency: EL PASO-RAVENNA 500 kV Line
Loading: 100.3% (Rating: 990 MVA)
DFAX: 5.0%
Pre-project loading: 97.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BLUEBONNET-JONES CHAPEL 230 kV	Rebuild line with higher-capacity conductor	\$32,091,000	9% (\$2,840,000)
MCNEIL-BREED 230 kV L	Rebuild line with higher-capacity conductor	\$15,264,000	5% (\$728,000)
SLOCUM RD SUBSTATION-MCNEIL 230 kV line	Rebuild line with higher-capacity conductor	\$8,855,000	5% (\$422,000)
SLOCUM RD-DEKALB 230 kV Line	Rebuild line with higher-capacity conductor	\$53,985,000	5% (\$2,573,000)
SIDNEY A MURRAY JR HYDROELECTRIC-24TH & GARDENDALE 230 kV	Reconductor line	\$8,552,000	5% (\$431,000)
RICHLAND SUBSTATION 500/230kV Transformer	Replace transformer with higher-rated unit	\$43,625,000	5% (\$2,150,000)
MINNEHAHA-NAPOLEON PEAKING STA SUBSTATION 230 kV	Reconductor line	\$24,320,000	5% (\$1,212,000)
RITTER PARK-SUGAR HILL 345 kV	Reconductor line	\$6,233,000	3% (\$184,000)
DEKALB SUBSTATION-RICHLAND 500 kV	Reconductor line	\$67,497,000	5% (\$3,257,000)
LIGHTHOUSE HILL-STEWART MTN DAM 345 kV Line	Reconductor line	\$8,834,000	3% (\$265,000)
DEKALB 500 kV/1 kV TX	Replace transformer with higher-rated unit	\$16,676,000	5% (\$760,000)
DEKALB 230/1kV TX	Replace transformer with higher-rated unit	\$7,977,000	5% (\$364,000)
ATASCOCITA-MAPLE STREET 230 kV	Reconductor line	\$15,527,000	6% (\$977,000)
LEETONIA-MAPLE STREET 230 kV	Reconductor line	\$6,633,000	6% (\$417,000)
	Total allocated to project		\$16,580,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.